

A Push-Pull Network Mechanism Revealed by Describing Function Analysis for Alzheimer's Pathological Oscillations

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Abstract—Abnormal theta band (4–8 Hz) cortical oscillations have been identified as a pathological hallmark in Alzheimer's disease (AD), which can be suppressed through high-frequency electrical stimulation (ES). However, their neural origins and mechanistic underpinnings remain poorly understood. In this work, we develop a mathematical model to characterize AD-related cortical theta oscillations. Through linear equivalent transformation of the physiological model and describing function analysis, we elucidate the network mechanism underlying these pathological oscillations. The results demonstrate that the reduced synaptic inhibition from fast inhibitory interneurons to pyramidal neuron populations drives the cortical network into a pathological low-frequency (theta band) oscillatory state. The oscillation frequency and amplitude can be directly determined from the intersection points between the Nyquist curve of the model's linear component and the negative inverse curve of the nonlinear element's describing function. Conversely, the separation between these curves driven by external stimulation visually represents how stimulation can shift the cortical network away from abnormal theta oscillations. The describing function analysis not only provides a mechanistic explanation for aberrant oscillations but also enables precise prediction of effective stimulation parameters for suppressing pathological theta activity. This work offers a novel framework to reveal the “push-pull” network effects of AD, providing an intuitive understanding of cortical stimulation mechanisms and a quantitative approach for optimizing neuromodulation strategies in AD.

Index Terms—Alzheimer's disease (AD), describing function, neural mass model (NMM), pathological oscillations, virtual cortical stimulation.

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I. INTRODUCTION

THE brain is the most complex organ of the human body [1]. The cerebral cortex, the brain region with the largest superficial area, plays a pivotal role in consciousness, memory, and cognition [2]. However, damage to certain specific brain regions can lead to neurological disorders such as Alzheimer's disease (AD). As a classic neurodegenerative disorder, in its early stages, the symptoms of AD manifest as a progressive loss of cognitive abilities [3] and gradually develop into massive neuronal degeneration in specific brain regions, including the hippocampus and temporal cortex [4].

While studies have been conducted on an electroencephalogram (EEG)/fMRI signal interpretation and AD stage classification [5], [6], [7], [8], the neurophysiological basis of rhythmopathies remains elusive. Compelling evidence from multiple studies confirms that aberrant theta band augmentation represents a typical pathological signature in AD patients [9], [10], [11]. Due to the difficulty in early diagnosis, AD treatment lags behind. This research shows that the disturbance of the excitation–inhibition balance in the cortex during the early stages of AD can cause changes in theta-band oscillations, which may serve as a biomarker for early diagnosis of AD [12], [13], [14].

Studies have shown that synaptic loss in AD patients is closely related to cognitive decline, indicating that synaptic dysfunction plays a causal role in the pathogenesis of AD [15], [16], [17], [18], [19], [20], [21], [22], [23], and the degree of synaptic degeneration has been shown to correlate positively with the severity of dementia [18], [26], [27]. Research in rodent models further supports this notion, indicating that accumulation of pathological tau triggers synaptic dysregulation and loss without causing extensive neuronal loss [24], [25]. Given that synaptic function and its variability are fundamental to the construction of functional brain networks [28], [29], exploring how synaptic alterations influence cortical network dynamics could provide critical insights into early-stage AD mechanisms.

For a deeper exploration of the underlying mechanisms of cerebral neuron activity, several computational modeling methodologies have emerged. Among them, the neural mass model (NMM) has gained popularity [30], [31], [32]. The NMM, composed of cortical columns or hippocampus, has shown excellent performance in simulating normal

physiological brain rhythmic activity and pathological brain activity in disease states and has been applied in various neurological fields such as epilepsy, cognitive impairment, and dementia. Jansen and Rit implemented the construction of NMMs for cortical columns and determined anatomically acceptable values for various model parameters [33], [34]. In order to study epilepsy, Wendling et al. [35] introduced a population of γ -aminobutyric acid (GABA) interneurons with fast synaptic kinetics and generated an EEG signal that closely approximated the physiological data. The development and usage of these models were reviewed by Fietkiewicz and Loparo [36]. Recently, Ranasinghe et al. [37] examined excitatory and inhibitory parameters of neuronal subpopulations by constructing an NMM and explored the relationship between amyloid-beta and tau deposition and AD neural oscillations.

Therefore, this work uses NMM to build a virtual cortical network including pyramidal neurons, excitatory interneurons, and slow/fast inhibitory interneurons, so as to reveal the underlying mechanistic factors that can push the network into a theta band pathological state of AD.

Extensive case studies and clinical trials have been undertaken to determine the potential of electrical stimulation (ES) as a treatment modality for patients with AD [38], [39], [40], [41]. Despite the benefits of ES for AD, the underlying mechanisms are poorly understood due to the complexity of the brain and ethical limitations. Attempts have been made to model and analyze the results of stimulation at different scales of the brain [42]; however, the determination of the optimal stimulation frequency, intensity, and target is still mainly dependent on experience. A computational model that can predict the parameter space favorable to oscillation inhibition may provide valuable guidance for clinical applications.

In this study, the mechanism of ES is explored by introducing virtual cortical stimulation in a model simulation. It is hypothesized that ES is able to pull the network away from the pathological state. The push-pull dynamics of virtual cortical stimulation for AD oscillations are visualized employing the describing function method [43], [44], [45], [46]. The describing function is used to linearize the nonlinear elements in the NMM, resulting in a cortical NMM that is input-output equivalent to a linear closed-loop system. The describing function method provides a way to detect the occurrence and disappearance of pathological oscillations by examining whether there is an intersection between the Nyquist curve (linear part) and the negative describing function curve (nonlinear part), which reveals how the push-pull mechanism can induce and suppress pathological oscillations in the cortical network and provides a prediction of the effective ES parameters that suppress the pathological theta oscillations.

The subsequent sections of this article are outlined as follows. Section II introduces the virtual cortical network physiological model and the equivalent simplified model based on the descriptive function analysis method. Section III performs the validation of the virtual cortical network, exploring how changes in synaptic action can push the network activity toward a pathological state (theta oscillations) and how ES pulls the network away from the pathological state; by utilizing the describing function method, the visualization of push-pull

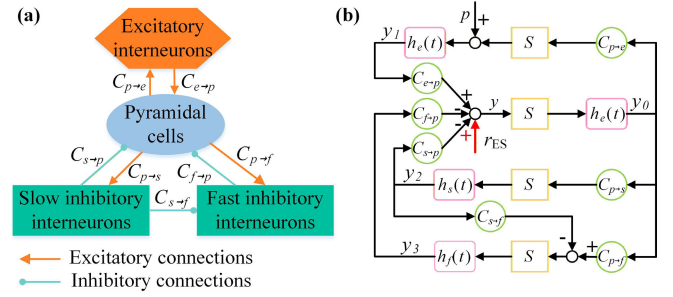


Fig. 1. Model of the cortical network. (a) Structure. (b) Schematic of the virtual cortical network with each population consisting of two components: the sigmoid function and the second-order impulse response functions.

effects is achieved. Finally, the discussions and conclusions are given in Section IV.

II. MODELS AND METHODS

A. Physiological Model of Alzheimer's Oscillations

Fig. 1(a) depicts a virtual cortical network, comprising four neural populations: pyramidal cells, excitatory interneurons, inhibitory interneurons with slow synaptic kinetics, and inhibitory interneurons with fast synaptic kinetics. For convenience, in the following, we will use p , e , s , and f to represent the four different neuron populations mentioned above, respectively. Pyramidal cells receive excitatory input from excitatory interneurons, and excitatory interneurons receive excitatory feedback from pyramidal cells. Meanwhile, pyramidal cells receive inhibitory input from slow and fast inhibitory interneurons; slow and fast inhibitory interneurons receive excitatory feedback from pyramidal cells; and fast inhibitory interneurons receive inhibitory input from slow inhibitory interneurons. To signify the respective coupling intensities within the network provided in Fig. 1(a), seven constants ($C_{i \rightarrow j}$) are employed. In our work, $p(t)$ (modeled by Gaussian white noise with a mean of 1.0 and a variance of 1.0) represents the average pulse density from adjacent or distant regions, and ES [i.e., r_{ES} in Fig. 1(b)] is set as a biphasic periodic pulse sequence, featuring equal durations and amplitudes for both its positive and negative phases.

Fig. 1(b) presents the detailed schematic of the NMM, where each constituent population incorporates a sigmoid function S and a second-order transfer function $G(s)$. The postsynaptic membrane potential is transformed into a presynaptic pulse density by the function S . The second-order transfer function, including excitatory, slow inhibitory, and fast inhibitory impulse response functions, represented by $h_e(t)$, $h_s(t)$, and $h_f(t)$, transforms the presynaptic impulse densities into corresponding postsynaptic membrane potentials.

Specifically, the function $S(v)$ can be described by the following equation:

$$S(v) = \frac{2e_0}{1 + e^{-rv}} - e_0 = e_0 \frac{1 - e^{-rv}}{1 + e^{-rv}} = e_0 \tanh\left(\frac{r}{2}v\right) \quad (1)$$

where $2e_0$ denotes the maximum firing rate of the neuronal population, r represents the slope of the functions $S(v)$, and v is the postsynaptic membrane potential.

The impulse response of the excitatory or inhibitory second-order transfer function can be described as

$$h_e(t) = Aate^{-at}, \quad h_s(t) = Bbte^{-bt}, \quad h_f(t) = Ggte^{-gt} \quad (2)$$

for $t \geq 0$, A represents the excitatory synaptic gain, B and G represent the slow and fast inhibitory synaptic gains, respectively, and a , b , and g are the synaptic time constants.

For an input/output system with input $x(t)$ and output $y(t)$, the transfer function is the ratio of the Laplace transform of the output $[Y(s)]$ to the Laplace transform of the input $[X(s)]$. For example, $G_e(s) = Y_e(s)/X_e(s) = A/a/(s/a + 1)^2$. Then, this transfer function can be directly related to the differential equation that defines the dynamic behavior of the input-output system in the time domain. From this, we obtain $s^2Y_e(s)/a + 2sY_e(s)/a + Y_e(s) = AX_e(s)/a$ as equivalent representation. By utilizing the properties of the Laplace transform, the time-domain differential equation is obtained, i.e., $\ddot{y}_e/a^2 + 2\dot{y}_e/a + y_e = Ax_e/a$, or $\ddot{y}_e = Aax_e - 2a\dot{y}_e - a^2y_e$. Thus, the dynamics of the NMM can be modeled by the following system of different equations:

$$\begin{aligned} \ddot{y}_0 &= AaS (C_{e \rightarrow p}y_1 - C_{s \rightarrow p}y_2 - C_{f \rightarrow p}y_3) - 2a\dot{y}_0 - a^2y_0 \\ \ddot{y}_1 &= Aa(p + S(C_{p \rightarrow e}y_0)) - 2a\dot{y}_1 - a^2y_1 \\ \ddot{y}_2 &= BbS(C_{p \rightarrow s}y_0) - 2b\dot{y}_2 - b^2y_2 \\ \ddot{y}_3 &= GgS(C_{p \rightarrow f}y_0 - C_{s \rightarrow f}y_2) - 2g\dot{y}_3 - g^2y_3. \end{aligned} \quad (3)$$

Numerical integration using Euler's method is used for computing the time-domain response of the NMM. To ensure numerical accuracy, the integration time step is set to 0.1 ms. Meanwhile, in the simulation experiments, the results of the first 4 s of the time-domain response are discarded to reduce the impact of the transients on the steady-state system response. The output of the NMM system $y(t)$ can be obtained as $y(t) = C_{e \rightarrow p}y_1(t) - C_{s \rightarrow p}y_2(t) - C_{f \rightarrow p}y_3(t)$, which represents the sum of the average postsynaptic potentials of the pyramidal cells.

To facilitate modeling, we selected a set of parameter values listed in Table I. These parameter values were selected and adjusted within established biophysical ranges provided by Wendling et al. [35], Fietkiewicz and Loparo [36], David et al. [47], and Touboul et al. [48] to ensure that the model, under our modified structure, could exhibit key neurodynamic features relevant to AD, particularly the pathological theta-band oscillations investigated in this study.

Considering that the loss of synapses in AD may disturb the excitation/inhibition balance and result in cognitive decline [40], the fast inhibitory loop plays a crucial role in the precision of pyramidal cell firing and the synchronization of low-frequency oscillations. A weakening of inhibitory synaptic coupling can lead to disinhibition of pyramidal neurons, excessive network excitation, and potentially abnormal oscillations of the theta band [33]. Specifically, fast inhibitory interneurons, particularly parvalbumin-positive (PV+) GABAergic neurons, are thought to weaken the inhibitory control of pyramidal neurons and lead to excessive excitation and slowed rhythm of the network [49]. Therefore, in simulating a pathological state, this study reduces the inhibitory synaptic strength

TABLE I
INTERPRETATION AND STANDARD VALUES OF NMM PARAMETERS

Parameters	Interpretation	Values
A	Average excitatory synaptic gain	3.25 mV
B	Average slow inhibitory synaptic gain	22 mV
G	Average fast inhibitory synaptic gain	10 mV
a	Average time constants of excitatory populations	100 s ⁻¹
b	Average time constants of slow inhibitory populations	50 s ⁻¹
g	Average time constants of fast inhibitory populations	200 s ⁻¹
$2e_0$	Maximum firing rate	5 s ⁻¹
r	Slope of nonlinear sigmoid function	0.56 mV ⁻¹
$C_{p \rightarrow e}$		$C=135$
$C_{e \rightarrow p}$		0.6C
$C_{p \rightarrow s}$		0.25C
$C_{s \rightarrow p}$	Coupling strength between populations $i \rightarrow j$	0.25C
$C_{p \rightarrow f}$		0.3C
$C_{s \rightarrow f}$		0.1C
$C_{f \rightarrow p}$		30(Normal) →20(AD)

$C_{f \rightarrow p}$ between fast inhibitory interneurons and pyramidal neurons to simulate the pathological state.

These parameter settings enable our model to produce both normal and AD states when the inhibitory synaptic strength $C_{f \rightarrow p}$ varies.

B. Describing Function Method and Equivalent Model

Synaptic dysfunction can induce pathological rhythmic activity in neural networks. Although external stimulation can modulate this abnormal state [50], the quantitative relationships between synaptic alterations and rhythmic activity characteristics, as well as between stimulation parameters and regulatory effects, remain not fully elucidated. Clarifying these relationships is crucial for understanding neuromodulation mechanisms and optimizing stimulation strategies. Notably, the frequency and amplitude of neural oscillations serve as key indicators of disease severity. In linear systems, the Nyquist stability criterion serves as a frequency-domain analysis method that provides intuitive visualization of critical oscillatory characteristics, including amplitude and frequency. However, our cortical network inherently constitutes a strongly nonlinear dynamical system due to its nonlinear activation functions (e.g., the sigmoid function) and multilayered cascaded structure. Thus, the describing function method, as an effective tool for analyzing the stability of nonlinear feedback systems, can be employed to determine whether such systems exhibit self-sustained oscillations [51], [52]. By computing the

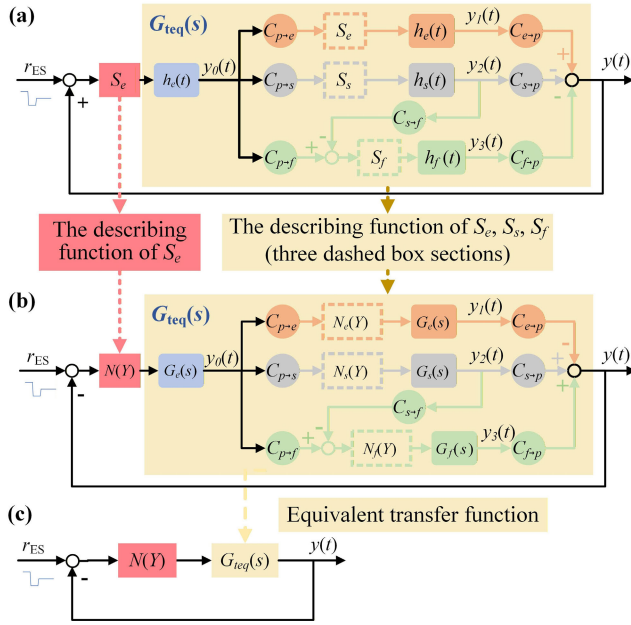


Fig. 2. Equivalent representation of the NMM. (a) Simplified illustration of the NMM in Fig. 1(b). (b) Restatement of part (a), equating the nonlinear function $S(v)$ with the describing functions $N_e(Y)$, $N_s(Y)$, and $N_f(Y)$. (c) Block diagram of the nonlinear system after equivalent linearization using the describing function method.

fundamental harmonic gain of a nonlinear element in response to a sinusoidal input, the nonlinearity can be equivalently represented as a linear component. This allows the application of the classical Nyquist stability criterion to nonlinear systems such as brain networks.

In this work, the describing function method serves as a tool for assessing the stability of nonlinear systems, specifically to investigate: 1) the impact of synaptic changes on pushing the virtual cortical network toward pathological states and 2) how ES pulls the network out of the pathological state. A visualization of this push-pull effect related to the pathological network oscillation can be provided by the describing function method, so as to intuitively predict the performance of ES at different parameter settings and obtain the optimal ES parameter subspace for a given AD (pathological) state. This method of rapidly tuning effective ES parameters may offer guidance for both experimental and clinical trials.

In this section, we adopt the describing function analysis method to transform the nonlinear NMM into a standard closed-loop structure consisting of a single nonlinear element and a linear component. By deriving the describing function expression of the nonlinear element, we achieve linearization of the entire system, as shown in Fig. 2.

The specific steps are given as follows.

Ideally assuming the input of the sigmoid function $S(v)$ as a single-frequency periodic signal $Y \sin(\omega v)$, where Y is the amplitude of the cortical oscillation and ω is the corresponding frequency, we can apply harmonic analysis to the steady-state output of $S(v)$. Generally, $S(v)$ is a nonsine periodic signal, so we can expand it in a Fourier series

$$S(v) = A_0 + \sum_{k=1}^{\infty} (A_k \cos(k\omega v) + B_k \sin(k\omega v)) \quad (4)$$

where A_0 denotes the direct current component and $A_k \cos(k\omega v) + B_k \sin(k\omega v)$ is the k_{th} harmonic component.

Because $S(v)$ is an odd symmetric function, the component A_0 is 0, and the Fourier coefficients A_k are 0 for all k . Given that the linear elements are low-pass, higher harmonics ($k > 1$) will be attenuated through the linear elements, so the first harmonic ($k = 1$) in the Fourier series of $S(v)$ is of interest, where $S(v) = B_1 \sin(\omega v)$. The coefficient B_1 is derived from the Fourier integral

$$B_1 = \frac{1}{\pi} \int_0^{2\pi} S(v) \sin(\omega v) d(\omega v). \quad (5)$$

By setting $x = \omega v$ and combining (1), we can obtain

$$\begin{aligned} B_1 &= \frac{1}{\pi} \int_0^{2\pi} S(Y \sin x) \sin x dx \\ &= \frac{e_0}{\pi} \int_0^{2\pi} \tanh\left(\frac{r}{2} Y \sin x\right) \sin x dx. \end{aligned} \quad (6)$$

For sinusoidal inputs, the complex ratio of the fundamental frequency harmonic constituent in the steady-state output to the input signal is the descriptive function of the nonlinear component. Therefore, when applying the describing function method to nonlinear control systems with linear low-pass elements, we focus on the dominant fundamental frequency harmonic component in the steady-state output of the nonlinear components. The describing function is defined as the complex ratio of the fundamental output harmonic to the input

$$N(Y) = \frac{B_1}{Y} = \frac{e_0}{\pi Y} \int_0^{2\pi} \tanh\left(\frac{r}{2} Y \sin x\right) \sin x dx. \quad (7)$$

When ES is applied to the model, the sigmoid function becomes the following equation:

$$S(v) = e_0 \left[\frac{\tau}{T} \tanh\left(\frac{r}{2}(v+m)\right) + \frac{\tau}{T} \tanh\left(\frac{r}{2}(v-m)\right) + \left(1 - \frac{2\tau}{T}\right) \tanh\left(\frac{r}{2}v\right) \right] \quad (8)$$

where τ is the duration of one ES cycle and m is the corresponding amplitude [43]. By carefully examining the role of τ in stimulation effects (see supplementary material), we set $\tau = 1$ ms, which is consistent with [53] and [54]. Then, the describing function analysis technique is applied to (8), so as to obtain the “equivalent nonlinearity.” Because the frequency of the stimulus is much higher than that of the local field potentials of the neural system, we can assume that the output $y(t)$ of the NMM remains constant within a stimulation period T . As a result, we can explore the response of the NMM under ES, i.e., to determine the effect of ES on the pathological oscillatory state of the cortical network.

The describing function $N(Y)$ of the nonlinear element $S(Y)$, including ES, is then given as

$$N(Y) = \frac{e_0}{\pi Y} \int_0^{2\pi} \left[\begin{aligned} &\frac{\tau}{T} \tanh\left(\frac{r}{2}(Y \sin x + m)\right) \\ &+ \frac{\tau}{T} \tanh\left(\frac{r}{2}(Y \sin x - m)\right) \\ &+ \left(1 - \frac{2\tau}{T}\right) \tanh\left(\frac{r}{2}Y \sin x\right) \end{aligned} \right] \sin x dx. \quad (9)$$

Based on this, the NMM can be further transformed into a simpler version. First, the nonlinear part of the classical closed-loop structure [red box in Fig. 2(a)] is replaced by the describing function as $N(Y)$. Second, the linear characteristics, denoted as $G_{\text{teq}}(s)$, are provided by the equivalent transfer function, as shown in the yellow shaded part in Fig. 2. The nonlinear elements $S(v)$ (three dashed boxes in Fig. 2(a), S_e , S_s , and S_f) are also represented by the describing function $N(Y)$ (three dashed box sections in Fig. 2(b), N_e , N_s , and N_f), which can further be expressed as the equivalent gain:

$$K_e = N_e(C_{p \rightarrow e} Y_0), \quad K_s = N_s(C_{p \rightarrow s} Y_0) \\ K_f = N_f(C_{p \rightarrow f} Y_1).$$

Here, Y_0 is the amplitude of $y_0(t)$, and Y_1 is the amplitude of $C_{p \rightarrow f} y_0(t) - C_{s \rightarrow f} y_2(t)$. Thus, $G_{\text{teq}}(s)$ can be written as

$$G_{\text{teq}}(s) = G_e(s) \\ \times \left(\begin{aligned} &C_{p \rightarrow s} K_s G_s(s) C_{s \rightarrow p} \\ &+ (C_{p \rightarrow f} - C_{p \rightarrow s} K_s G_s(s) C_{s \rightarrow f}) K_f G_f(s) C_{f \rightarrow p} \\ &- C_{p \rightarrow e} K_e G_e(s) C_{e \rightarrow p} \end{aligned} \right). \quad (10)$$

Here, excitatory and inhibitory roles are distinguished by positive and negative signs in (3) and (10). Despite equivalent simplification, the model can still capture the core pathological mechanism of excitation–inhibition imbalance in AD. The frequency-domain transfer function of the (stable) equivalent transfer function $G_{\text{teq}}(s)$ can be recast as $G_{\text{teq}}(j\omega)$.

Based on the above analysis, we use the describing function method to equate the NMM of the virtual cortical network and convert it into a classical closed-loop system with characteristic equation $N(Y)G_{\text{teq}}(s) + 1 = 0$, as illustrated in Fig. 2(c), where $N(Y)$ indicates the describing function of the nonlinear element of NMM, while $G_{\text{teq}}(s)$ denotes the transfer function of the linear part.

The amplitude Y and frequency ω of oscillations in this nonlinear system can be calculated by solving the characteristic equation $N(Y)G_{\text{teq}}(s) + 1 = 0$ of the equivalent cortical network. The negative inverse curve $-1/N(Y)$ of the nonlinear part and the Nyquist curve of the linear part $G_{\text{teq}}(j\omega)$ can be coexpressed in the complex s plane. The Nyquist curve is plotted on the complex s plane with frequency ω as the parameter, at which point we can analyze the stability of this nonlinear system in three cases.

- 1) The NMM system is asymptotically stable when the Nyquist curve of $G_{\text{teq}}(j\omega)$ does not encircle the $-1/N(Y)$ curve.
- 2) The NMM system is unstable when the Nyquist curve of $G_{\text{teq}}(j\omega)$ encircles the $-1/N(Y)$ curve.

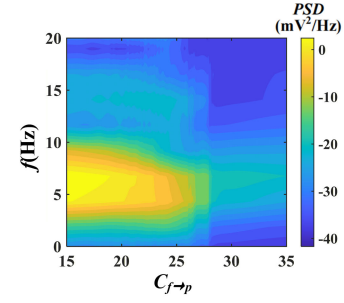


Fig. 3. Dependence of the PSD on the synaptic gain $C_{f \rightarrow p}$.

- 3) If the two curves intersect at a point, the NMM system is marginally stable, and periodic oscillations will occur.

Because the NMM nonlinear system can exhibit spontaneous oscillations, there may be intersection points of the two curves, so we need to make further judgments about the system.

- 1) If the $-1/N(Y)$ curve enters the unstable region from the stable region along the direction of increasing amplitude, the intersection point is an unstable periodic motion.
- 2) If the $-1/N(Y)$ curve enters the stable region from the unstable region, then the intersection point is a stable periodic motion, i.e., the NMM exhibits stable spontaneous oscillation, and the corresponding amplitude Y at the intersection point is the amplitude of the system oscillation.

Since AD has a stable rhythmic characteristic [i.e., satisfying condition (2) above], we consider using describing function analysis to explore the mechanism underlying the oscillations and investigate the role of ES in suppressing the pathological oscillations.

III. RESULTS

This section focuses on applying the describing function method to the analysis of the push-pull mechanism, in order to determine a suitable set of ES parameters that are capable of alleviating pathological AD oscillations.

A. Reduced Synaptic Action Pushes Network Toward a Pathological Oscillatory State

Fig. 3 provides an illustration of the relationship between fast inhibitory synaptic gain and the peak frequency of the power spectral density (PSD).

By keeping all other parameters at standard values and varying only the fast inhibitory synaptic gain $C_{f \rightarrow p}$ within the range of 15–35, it is observed that as $C_{f \rightarrow p}$ decreases, cortical network oscillations gradually concentrate in the theta band, accompanied by an increase in the PSD peak.

According to the classic NMM, we can achieve a normal physiological state when $C_{f \rightarrow p}$ is set to 30. Previous AD EEG studies show that, compared to healthy individuals, AD patients exhibit an increase of approximately 50%–100% in theta-band power [55], [56], [57]. As shown in Fig. 3, reducing $C_{f \rightarrow p}$ below the normal value enhances theta-band activity and drives the system toward a pathological oscillatory state. Based

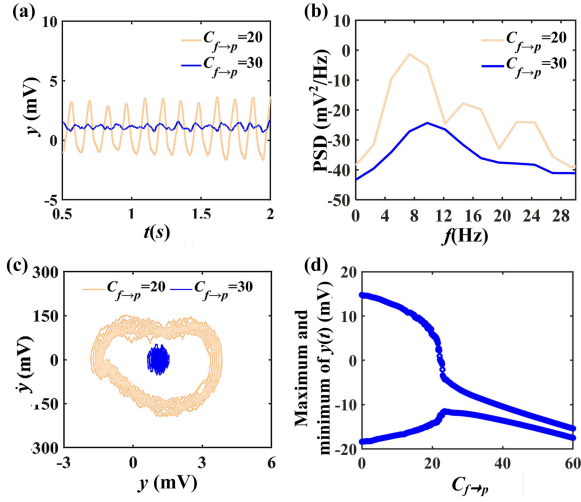


Fig. 4. Comparison of pathological ($C_{f \rightarrow p} = 20$) and normal physiological ($C_{f \rightarrow p} = 30$) states of the virtual cortical network. (a) EEG signal. (b) Power spectrum. (c) Phase trajectory. (d) Extreme values (maximum and minimum) of the cortical network under different coupling parameters $C_{f \rightarrow p}$.

on this parameter exploration, $C_{f \rightarrow p} = 20$ was selected as a representative pathological condition. At this value, the model exhibits stable and pronounced theta-band oscillations while remaining close to biologically plausible ranges.

Specifically, in Fig. 4(a), when $C_{f \rightarrow p}$ is adjusted from 30 to 20, the virtual cortical network transitions from normal weak oscillation activity to a pathological state characterized by excessive oscillation. Fig. 4(b) shows the related power spectrum, where the power density at the main frequency of pathological oscillations in the theta band significantly increases. In order to further explore the dynamic characteristics behind the activity changes, the corresponding phase trajectories are shown in Fig. 4(c). It can be seen that, when the value of $C_{f \rightarrow p}$ decreases, the stable equilibrium state (blue curve) changes to a new limit cycle attractor (orange curve), which indicates that the output activity of the NMM exhibits oscillations due to the loss of synapses. More specifically, as the value of $C_{f \rightarrow p}$ decreases, the limit cycle range enlarges. By reducing the synaptic strength of fast inhibitory interneurons on pyramidal neurons, the output activity of the virtual cortical network transitions from weak oscillatory activity, representing the normal state, to overoscillatory activity, representing the pathological state. As observed in Fig. 4(d), as $C_{f \rightarrow p}$ decreases, the disparity between the maximum and minimum outputs of the cortical network intensifies.

The reduction in $C_{f \rightarrow p}$ weakens fast inhibition onto pyramidal neurons, leading to their disinhibition and subsequent emergence of aberrant low-frequency oscillations in cortical circuits. This process can be accurately characterized through the describing function method. The identified mechanism not only clarifies the neurophysiological foundations of AD-related pathological oscillations but also establishes a virtual patient platform for ES parameter optimization. Thus, we can control these changes in virtual patients obtained by NMM to test the stimulation parameters to suppress abnormal theta rhythms in AD and achieve precise regulation.

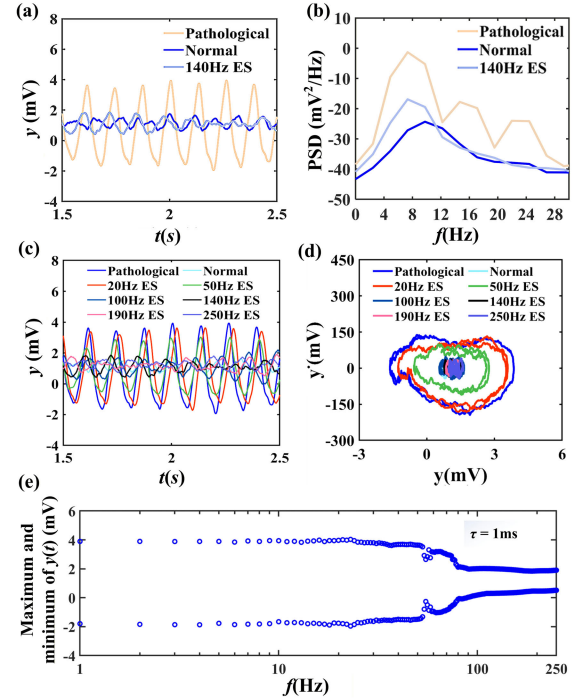


Fig. 5. Effect of virtual ES on cortical network oscillation. (a) EEG signals in normal, pathological state, and under ES conditions corresponding to (a). (b) PSD analysis corresponding to (a). (c) Outputs of the cortical network at different ES frequencies. (d) Phase trajectory diagram corresponding to (c). (e) Extreme values (maximum and minimum) of cortical network output at different ES frequencies.

B. ES Pulls the Network Away From the Pathological Oscillation

To investigate the impact of the change in synaptic action on the pathological state of the cortical network, we reduced the synaptic strength of fast inhibitory interneurons on pyramidal neurons. As shown in Section III-A, the virtual cortical network exhibits pathological oscillatory activity when the parameter $C_{f \rightarrow p}$ is reduced from 30 to 20. This change pushes the virtual cortical network toward pathological oscillatory activity. To investigate the impact of stimulation on oscillatory activity, we incorporate virtual cortical stimulation into our analysis. By adjusting the connection strength $C_{f \rightarrow p} = 20$, the network simulates a pathological state. The frequency of pulsatile biphasic ES introduced into our system is 130 Hz. By observing the output activity and power spectrum of the pathological system with and without ES, it can be seen that ES can successfully suppress the pathological oscillation, as shown in Fig. 5(a). Here, the ES pulse amplitude is configured to $m = 4$ mA, while the duration of each pulse is specified as $\tau = 1$ ms [53], [54].

To observe the dynamic characteristics of the cortical network under ES, we examine the effect of ES at different frequencies. The results are depicted in Fig. 5(c), and the corresponding phase trajectories are shown in Fig. 5(d). The phase trajectories of the cortical network output activity over a 1-s time period are shown in Fig. 5(d). ES utilizing frequencies exceeding 100 Hz can alleviate the pathological oscillations, as indicated by the disappearance of the limit cycle in Fig. 5(d)

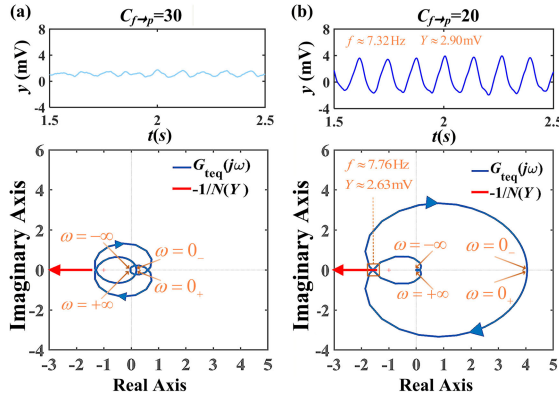


Fig. 6. Nyquist diagrams and model simulation results with different coupling parameters. (a) Normal system ($C_{f \rightarrow p} = 30$). (b) Pathological system ($C_{f \rightarrow p} = 20$).

and the shift to a new equilibrium state. This intuitively illustrates the impact of high-frequency ES on the pathological oscillations, i.e., high-frequency ES can pull the system into a new equilibrium state and suppress the pathological oscillations. Meanwhile, it can be seen from Fig. 5(d) that, relative to the light blue curve (normal state), the phase trajectory moves to the right in the horizontal direction under the action of 100 Hz (dark blue curve), 140 Hz (black curve), 190 Hz (pink curve), and 250 Hz (purple curve) ES, which exhibits the way of ES moving the system to a new equilibrium state. Low-frequency ES (e.g., $f = 20$ Hz and $f = 50$ Hz) added to the cortical network does not attenuate the pathological oscillations.

To further explore the influence of ES on the generation of new equilibrium states, Fig. 5(e) presents the bifurcation plot of the virtual cortical network's output. The bifurcation parameter employed in our analysis is the ES frequency f , and as f escalates to a comparably high level, the pathological network activity shifts to a new steady state. This suggests that high-frequency ES could pull the network away from a pathological state. With high-frequency ES, the cortical network reproduces a weak physiological state by effectively inhibiting the pathological oscillatory activity, which suggests the pull impact of the ES on the cortical dynamics.

C. Visualization of the Push-Pull Mechanism and Determination of the Relationship Between ES Parameters by Describing Function Analysis

The Nyquist curves of the virtual cortical network system without ES in the normal state ($C_{f \rightarrow p} = 30$) and the pathological state ($C_{f \rightarrow p} = 20$) are compared in Fig. 6.

The Nyquist curve, indicated by the blue curve, and the negative inverse curve, indicated by the red curve, do not intersect, suggesting that there are no oscillations occurring in the system. When the synaptic inhibition of fast inhibitory interneurons on pyramidal neurons is reduced, the range of the Nyquist curve (the blue curve) increases and intersects the negative inverse describing function curve (red arrow). From the data that correspond to the intersection, it can be determined that there is a 7.76-Hz oscillation with an

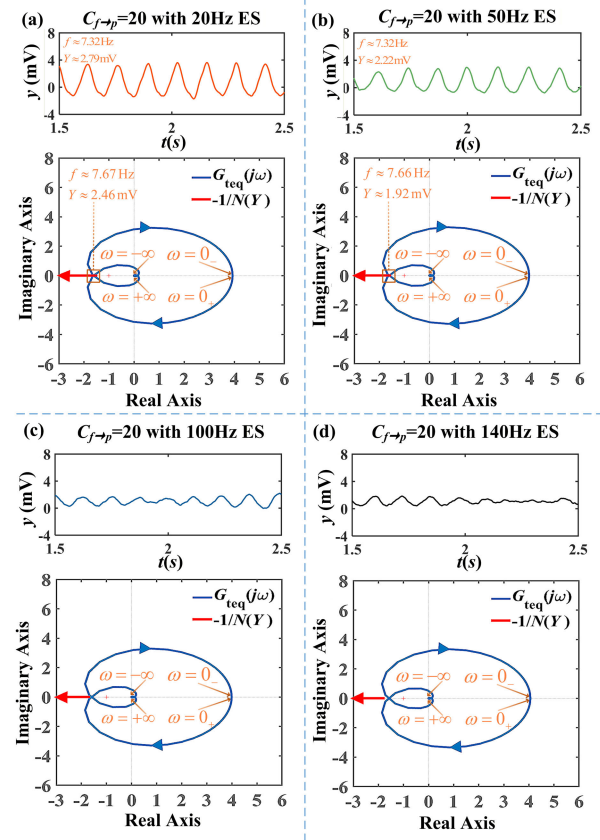


Fig. 7. (a)–(d) Influence of different frequencies of ES on the cortical network, utilizing the describing function analysis. The amplitude of ES is set to 4 mA, and the duration of each pulse is 1 ms.

amplitude of 2.63 mV that is predicted from this theoretical analysis. In contrast to the simulation outcomes presented in Fig. 4, the describing function analysis can illustrate the mechanism (reduction of the synaptic inhibition of fast inhibitory interneurons on pyramidal neurons) that can push the cortical network into a pathological state.

In order to reveal the ES mechanism that can pull the cortical network away from the pathological state, we carry out the describing function analysis for various ES with different frequencies.

In Fig. 7(c), high-frequency ES induces a left shift of the red negative inverse describing function, resulting in no intersection with the blue Nyquist curve of $G_{\text{teq}}(j\omega)$. The absence of an intersection indicates the disappearance of oscillations in the system, demonstrating the “pull” impact of the ES. By further increasing the stimulation frequency, it can be found that the starting point of the red negative inverse describing function shifts further to the left. Thus, from the perspective of the theoretical analysis, it is demonstrated that high-frequency ES can suppress the abnormal cortical oscillations [see Fig. 7(d)]. By adjusting the ES frequency to a low level, it can be easily predicted that an undesirable outcome may occur in the system. As shown in Fig. 7(a) and (b), with low-frequency ES ($f = 20$ Hz or $f = 50$ Hz), intersections of the two curves still exist, suggesting that low-frequency cortical ES fails to pull the cortical network away from the pathological state. The describing function analysis provides an efficient

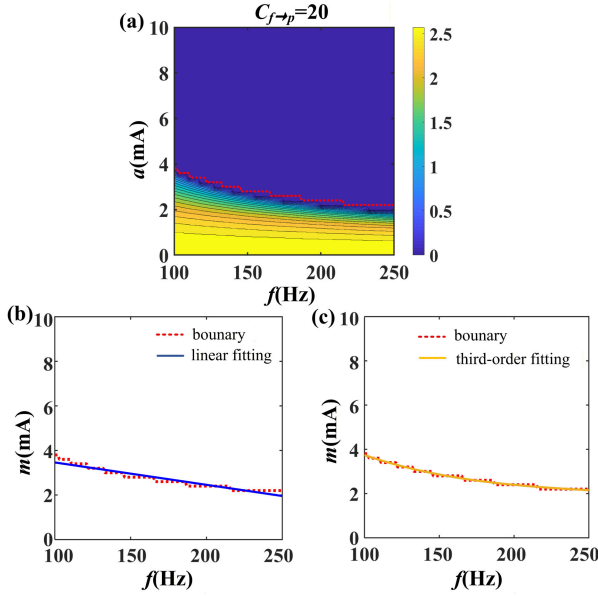


Fig. 8. (a) Combined effects of ES frequency and amplitude on virtual cortical network oscillations. The oscillation-suppressed and nonsuppressed regions obtained via the describing function method, along with the suppression boundary (red line) derived from cortical network simulations. (b) Linear and (c) third-order polynomial fitting of the suppressing boundary.

prediction of the significance of the cortical stimulation and provides a way to visually explore the mechanism by which ES can impact the pathological AD oscillations. In Figs. 6 and 7, the predicted results of the function analysis are consistent with the trend of the numerical simulation results of the model (see the top figures of Figs. 6 and 7).

By using the describing function analysis method, the range of ES parameters that can suppress pathological theta oscillations is predicted, as shown in Fig. 8. The depth of the color bar is related to the oscillation amplitude of the virtual cortical network, and the yellow part is the area where the oscillation cannot be suppressed, while the blue area is successfully suppressed.

Fig. 8 exhibits the collaborative impact of various ES amplitudes and frequencies with a pulse duration of 1 ms on the pathological theta oscillations. The red line depicts the suppression oscillation boundary calculated through computational simulations of the cortical model, using the same range of ES parameters. It can be observed that the simulated results are in agreement with the theoretical prediction by the describing function method, which also validates the effectiveness of the describing function analysis approach.

By applying the methods of curve-fitting, the suppressing boundaries in Figs. 8 and 9 can be described by linear expressions or third-order polynomials, which characterize the optimal relationship of stimulation frequency and amplitude to move the system into the “Oscillation Suppression” region

$$m = -0.01003f + 4.464 \quad (11)$$

$$m = -2.499 \times 10^{-7}f^3 + 1.948 \times 10^{-4}f^2 - 0.05441f + 7.482. \quad (12)$$

The results show that ES can pull the cortical network away from the pathological state under appropriate ES parameter

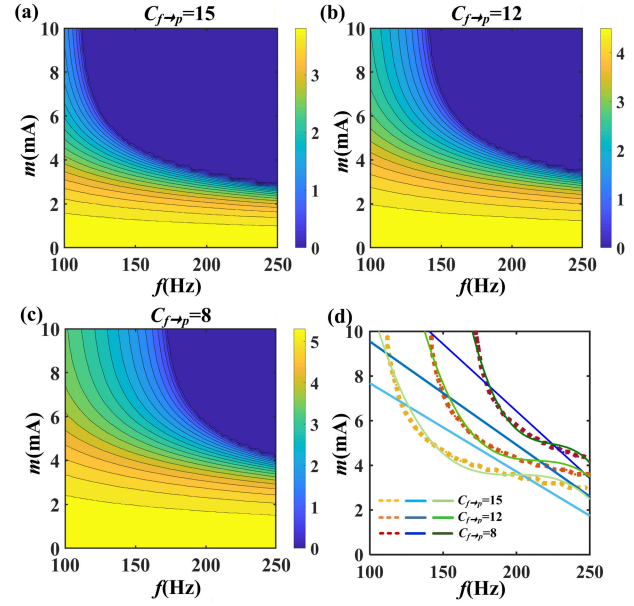


Fig. 9. Role of stimulation parameters on the pathological cortical network at various synaptic coupling strengths. (a) $C_{f \rightarrow p} = 15$. (b) $C_{f \rightarrow p} = 12$. (c) $C_{f \rightarrow p} = 8$. (d) Linear and third-order polynomial fitting of the suppressing boundary (dashed line).

settings. There is a clear relationship between the frequency and amplitude of ES for suppressing pathological oscillations, i.e., the higher the ES frequency, the lower the ES amplitude. The effective ES parameter range can be efficiently calculated from the describing function method. Thus, we can draw a conclusion that the describing function analysis method enables the visualization of the push-pull effects caused by virtual cortical stimulation on the pathological AD oscillations; furthermore, it can also provide a robust predictive capability for determining the optimal range of stimulation parameters necessary to effectively suppress the pathological theta oscillations.

To assess the effectiveness of the describing function analysis, we use it to predict the effective stimulation parameters of the pathological cortical network at various synaptic coupling strengths, such as $C_{f \rightarrow p} = 15$, $C_{f \rightarrow p} = 12$, and $C_{f \rightarrow p} = 8$, and the results are illustrated in Fig. 9(a)–(c). We find that the range of ES parameters that can effectively inhibit pathological oscillations changed significantly as the synaptic inhibitory effect of fast inhibitory interneurons on pyramidal neurons diminished. In Fig. 9(d), the boundary can be fit to the following expression.

First-Order Fitting:

$$C_{f \rightarrow p} = 15 : m = -0.03959f + 11.64 \quad (13)$$

$$C_{f \rightarrow p} = 12 : m = -0.04629f + 14.18 \quad (14)$$

$$C_{f \rightarrow p} = 8 : m = -0.06005f + 18.44. \quad (15)$$

Third-Order Fitting:

$$C_{f \rightarrow p} = 15 \\ m = -7.999 \times 10^{-6}f^3 + 4.772 \times 10^{-3}f^2 - 0.9494f + 66.57 \quad (16)$$

$$C_{f \rightarrow p} = 12$$

$$m = -1.237 \times 10^{-5} f^3 + 7.929 \times 10^{-3} f^2 - 1.698 f + 125.7 \quad (17)$$

$$C_{f \rightarrow p} = 8$$

$$m = -2.856 \times 10^{-5} f^3 + 0.01915 f^2 - 4.292 f + 326.5. \quad (18)$$

When the coupling strength $C_{f \rightarrow p}$ weakens (i.e., the AD oscillations worsen), the relationship curves go upper right, which indicates that the more severe pathological oscillations require stronger stimulation. We can quickly select effective ES values to suppress individual pathological oscillations via the describing function analysis method, which can provide guidance for the clinical application of ES to AD.

IV. DISCUSSION AND CONCLUSION

This study presents a cortical network model incorporating pyramidal cells along with excitatory, slow inhibitory, and fast inhibitory interneurons to investigate early-stage AD pathophysiology. Our framework demonstrates how synaptic dysfunction pushes cortical networks toward pathological states, while virtual cortical stimulation pulls the network away from the pathological state.

Specifically, the virtual cortical network in the classical closed-loop model is divided into nonlinear and linear parts by the describing function method. The results show that the coverage of the Nyquist curve increases as the synaptic inhibition of fast inhibitory interneurons on pyramidal neurons decreases, while the starting point of the negative inverse curve of the describing function remains unchanged. The occurrence of the intersection of these two curves indicates that the cortical network is pushed to the pathological state due to reduced synaptic inhibition.

The information of the intersection of the two curves determines the amplitude Y and frequency ω of the oscillations in the nonlinear cortical network, which is used to analyze the oscillatory characteristics of the nonlinear NMM system. The theoretical results reveal a possible mechanism for the development of early AD oscillations from a kinetic perspective.

Recent studies have successfully employed direct numerical simulations of NMM to explore parameter spaces and bifurcations in related contexts [58], [59]. Our work complements these efforts by proposing a linearization-based framework and leveraging the describing function method to rapidly explore the parameter space of stimulation frequency and amplitude, thereby enhancing computational efficiency. The describing function method offers a convenient framework for parameter space visualization and prediction, enabling systematic analysis of how stimulation amplitude and frequency modulate pathological AD oscillations. By leveraging this method to identify the minimal effective stimulation amplitude and optimal frequency, we can significantly reduce energy requirements reflected by the lower amplitude of the pulse ES while guiding the development of personalized ES therapies.

At the mechanistic level, existing literature generally attributes the EEG “slowing” in AD to excitation–inhibition imbalance, particularly resulting from impaired function of GABAergic fast interneurons. Building upon this foundation,

our model proposes a “push-pull” network mechanism that highlights the regulatory role of fast inhibition on pyramidal neurons and the consequent rhythm slowing following its loss, which aligns with recent experimental findings of PV-mediated inhibitory dysfunction leading to theta enhancement.

Although this work mainly focuses on the theoretical analysis and quick prediction of the external ES, it can be empirically validated by clinical data, such as our prior clinical EEG study [60], which utilized cubature Kalman filtering to estimate neural parameters from clinical EEG data. Key findings included significantly elevated theta frequency in AD patients, coupled with reduced cortical inhibitory neuron density, and weakened inhibitory synaptic coupling. These results align precisely with our current model’s mechanistic proposal: impaired fast inhibition underlies both local cortical hyperexcitability and theta-band synchronization in AD. Based on this, in subsequent studies, we could obtain patient-specific parameters from biological data and combine them with describing function-based stimulation optimization to enhance the biological reliability of our model and promote precision medicine in AD treatment.

In addition, in this work, we mainly explore the occurrence of pathological theta-band oscillations caused by synaptic inhibition defects in the cortical network. Although this method helps to understand a potential cause of oscillatory dysfunction in AD, it simplifies the overall picture of electrophysiological abnormalities observed in AD patients. It is worth noting that some empirical studies have shown that the decrease in power of the alpha band (8–12 Hz) and the beta band (13–30 Hz) is also a significant hallmark neurophysiological feature of AD [61], [62]. In the future, we consider more biological rhythms to verify the effectiveness of our method.

Third, external inputs in our work are modeled as simplified random noise rather than detailed projections from specific nuclei, which only capture a portion of the complex oscillatory changes in AD. Therefore, in future work, we will expand the current model, consider brain networks composed of more components, and simulate more comprehensive brain dynamics mechanisms to enhance biological fidelity.

In conclusion, this work presents a virtual cortical network to investigate the neural basis of pathological oscillation associated with AD and to reveal the mechanism underlying virtual cortical stimulation. The describing function analysis assists with visualizing the push-pull network effects and provides an intuitive visualization platform for selecting the appropriate stimulation parameters for a specific pathological rhythm, such as theta oscillation in the AD state. This work provides a novel perspective to develop effective stimulation for other neurological diseases, such as Parkinson’s disease and disorders of consciousness, based on a transparent dynamic mechanism.

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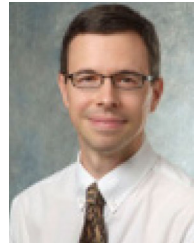
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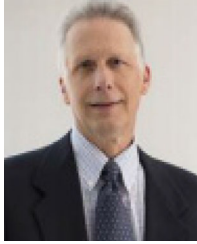
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